

The sharp lower-limit constant for the first decrease of a harmonic denominator

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Abstract

For positive integers $a \leq b$, write

$$\sum_{j=a}^b \frac{1}{j} = \frac{u_{a,b}}{v_{a,b}}, \quad \gcd(u_{a,b}, v_{a,b}) = 1, \quad v_{a,b} > 0,$$

and let $b(a)$ be the least $b > a$ for which $v_{a,b} < v_{a,b-1}$. For

$$f_d(X) = \sum_{i=0}^d \prod_{\substack{0 \leq j \leq d \\ j \neq i}} (X - j)$$

and for the density δ_d of primes modulo which f_d has a root, put

$$c = \sum_{d=1}^{\infty} \frac{\delta_d}{d(d+1)}.$$

We prove that, for every $\varepsilon > 0$, there are infinitely many pairs $a < b$ with

$$b < a + \left(\frac{1}{1+c} + \varepsilon \right) \log a \quad \text{and} \quad v_{a,b} < v_{a,b-1}.$$

Together with the reverse inequality established in the accompanying manuscript [1], this gives

$$\liminf_{a \rightarrow \infty} \frac{b(a) - a}{\log a} = \frac{1}{1+c}.$$

The main additional input is a finite-field form of Halász's inequality; we also use simultaneous Dirichlet approximation in an elementary auxiliary construction.

1 Notation and the main theorem

For $a \leq b$, define

$$L_{a,b} = \text{lcm}(a, a+1, \dots, b), \quad X_{a,b} = L_{a,b} \sum_{j=a}^b \frac{1}{j}, \quad g_{a,b} = \gcd(X_{a,b}, L_{a,b}).$$

Then

$$v_{a,b} = \frac{L_{a,b}}{g_{a,b}}.$$

We use ν_p for the p -adic valuation. All logarithms are natural.

For $d \geq 0$, let

$$f_d(X) = \sum_{i=0}^d \prod_{\substack{0 \leq j \leq d \\ j \neq i}} (X - j).$$

In particular, $f_0 = 1$. Whenever $x \notin \{0, 1, \dots, d\}$ in a field,

$$\sum_{i=0}^d \frac{1}{x-i} = \frac{f_d(x)}{\prod_{i=0}^d (x-i)}. \quad (1)$$

Let δ_d be the density of primes p for which f_d has a root modulo p . This density exists by Chebotarev's theorem [3]. Set

$$c = \sum_{d=1}^{\infty} \frac{\delta_d}{d(d+1)}. \quad (2)$$

Theorem 1. *For every $\varepsilon > 0$, there are infinitely many positive integers $a < b$ such that*

$$b < a + \left(\frac{1}{1+c} + \varepsilon \right) \log a$$

and

$$v_{a,b} < v_{a,b-1}.$$

Consequently,

$$\liminf_{a \rightarrow \infty} \frac{b(a) - a}{\log a} \leq \frac{1}{1+c}.$$

Corollary 2. *Combined with the lower bound from [1],*

$$\liminf_{a \rightarrow \infty} \frac{b(a) - a}{\log a} \geq \frac{1}{1+c},$$

Theorem 1 gives

$$\liminf_{a \rightarrow \infty} \frac{b(a) - a}{\log a} = \frac{1}{1+c}.$$

2 Local denominator calculations

We first isolate the elementary facts about adding the endpoint $1/b$.

Lemma 3. *Let U/V be a reduced fraction with $V > 0$, and let V' be the denominator in lowest terms of $U/V + 1/b$. Then, for every prime p ,*

$$\nu_p(V') - \nu_p(V) \leq \nu_p(b).$$

Proof. Put $s = \nu_p(V)$ and $k = \nu_p(b)$. Since $p \nmid U$,

$$\frac{U}{V} + \frac{1}{b} = \frac{bU + V}{bV}.$$

If $k < s$, then $\nu_p(bU + V) = k$, so the new denominator exponent is s . If $k > s$, then $\nu_p(bU + V) = s$, so the new denominator exponent is k . If $k = s$, the new denominator exponent is at most s . The assertion follows in all three cases. $\square$

Lemma 4. For every $d \geq 0$,

$$f_d(d - X) = (-1)^d f_d(X). \quad (3)$$

Moreover, if $p > d$ is prime and $f_d(x) = 0$ in $\mathbb{F}_p$, then

$$x \notin \{0, 1, \dots, d\}. \quad (4)$$

Proof. Replacing j by $d - j$ gives

$$\begin{aligned} f_d(d - X) &= \sum_{i=0}^d \prod_{\substack{0 \leq j \leq d \\ j \neq i}} (d - X - j) \\ &= (-1)^d \sum_{i=0}^d \prod_{\substack{0 \leq j \leq d \\ j \neq i}} (X - j) = (-1)^d f_d(X). \end{aligned}$$

If $x = i \in \{0, \dots, d\}$, every summand in $f_d(i)$ vanishes except the summand indexed by i , and

$$f_d(i) = \prod_{\substack{0 \leq j \leq d \\ j \neq i}} (i - j) \not\equiv 0 \pmod{p}$$

because $p > d$. □

The following two lemmas concern primes larger than the square root of the interval length.

Lemma 5 (A soluble prime gives a decrease). *Let $n = b - a$, let $\sqrt{n} < p \leq n$ be prime, and put $d = \lfloor n/p \rfloor$. Suppose that $p \mid b$ and, with*

$$z \equiv \frac{b}{p} \pmod{p},$$

we have $f_d(z) = 0$ in $\mathbb{F}_p$. Then

$$\nu_p(v_{a,b}) \leq \nu_p(v_{a,b-1}) - 1.$$

Proof. By Lemma 4, $z \notin \{0, \dots, d\}$. The multiples of p in $[a, b]$ are

$$b, b - p, \dots, b - dp.$$

For $0 \leq i \leq d$,

$$\frac{b - ip}{p} \equiv z - i \not\equiv 0 \pmod{p}.$$

Thus none of these integers is divisible by p^2 . Hence

$$\nu_p(L_{a,b}) = \nu_p(L_{a,b-1}) = 1.$$

Modulo p , only the displayed multiples of p contribute to $X_{a,b}$, and therefore, using (1),

$$\begin{aligned} X_{a,b} &\equiv \frac{L_{a,b}}{p} \sum_{i=0}^d \frac{1}{z - i} \\ &\equiv \frac{L_{a,b}}{p} \frac{f_d(z)}{\prod_{i=0}^d (z - i)} \equiv 0 \pmod{p}. \end{aligned}$$

On the other hand, $L_{a,b}/b$ is a p -adic unit. Since

$$X_{a,b} = \frac{L_{a,b}}{L_{a,b-1}} X_{a,b-1} + \frac{L_{a,b}}{b}$$

and $L_{a,b}/L_{a,b-1}$ is also a p -adic unit, it follows that $p \nmid X_{a,b-1}$. Thus the old denominator contains one factor p , while the new denominator contains none. $\square$

Lemma 6 (The bad residues for an insoluble prime). *Let $n = b - a$, let $\sqrt{n} < p \leq n$ be prime, and put $d = \lfloor n/p \rfloor$. Suppose that $p \mid b$ and that f_d has no root in $\mathbb{F}_p$. Define*

$$\mathcal{Z}_{p,d} = \{0\} \cup \{z \in \mathbb{F}_p \setminus \{0, 1, \dots, d\} : f_{d-1}(z-1) = 0\}. \quad (5)$$

If

$$z \equiv \frac{b}{p} \pmod{p} \quad \text{satisfies} \quad z \notin \mathcal{Z}_{p,d},$$

then

$$\nu_p(v_{a,b}) \leq \nu_p(v_{a,b-1}).$$

Moreover,

$$|\mathcal{Z}_{p,d}| \leq d. \quad (6)$$

Proof. First suppose that $z \in \{1, \dots, d\}$. Then $b - zp$ is divisible by p^2 . Since $p^2 > n$, it is the unique multiple of p^2 in $[a, b]$, and it belongs to both $[a, b-1]$ and $[a, b]$. If $p^k \parallel b - zp$, then this is the unique term of maximal p -adic order in both intervals. After multiplication by the corresponding least common multiple, that term is a p -adic unit and every other term is divisible by p . Hence both $X_{a,b-1}$ and $X_{a,b}$ are nonzero modulo p , and the p -parts of the two reduced denominators are equal.

Now suppose that $z \notin \{0, 1, \dots, d\}$. Then none of

$$b, b-p, \dots, b-dp$$

is divisible by p^2 , and

$$\nu_p(L_{a,b}) = \nu_p(L_{a,b-1}) = 1.$$

Since f_d has no root modulo p , equation (1) gives

$$X_{a,b} \equiv \frac{L_{a,b}}{p} \frac{f_d(z)}{\prod_{i=0}^d (z-i)} \not\equiv 0 \pmod{p}.$$

Similarly,

$$\begin{aligned} X_{a,b-1} &\equiv \frac{L_{a,b-1}}{p} \sum_{i=1}^d \frac{1}{z-i} \\ &\equiv \frac{L_{a,b-1}}{p} \frac{f_{d-1}(z-1)}{\prod_{i=1}^d (z-i)} \not\equiv 0 \pmod{p}, \end{aligned}$$

because $z \notin \mathcal{Z}_{p,d}$. Again the two p -parts are equal.

Finally, f_{d-1} has degree $d-1$ and is not the zero polynomial modulo p , since its leading coefficient is d and $p > d$. Hence the second set in (5) has at most $d-1$ elements, proving (6). $\square$

We shall also use a fixed collection of small primes, but at growing prime-power scales.

Lemma 7 (A prime-power-scale decrease). *Let $\ell > 3$ be prime, let $m = \ell^e$, and suppose*

$$2m \leq n = b - a < 3m, \quad m \mid b.$$

Put $z = b/m$. If $f_2(z) = 0$ in $\mathbb{F}_\ell$, then

$$\nu_\ell(v_{a,b}) \leq \nu_\ell(v_{a,b-1}) - 1.$$

Proof. The multiples of m in $[a, b]$ are exactly

$$b, b - m, b - 2m.$$

By Lemma 4, $z, z - 1, z - 2$ are nonzero modulo ℓ . Thus none of these three integers is divisible by ℓ^{e+1} , and

$$\nu_\ell(L_{a,b}) = \nu_\ell(L_{a,b-1}) = e.$$

Modulo ℓ ,

$$\begin{aligned} X_{a,b} &\equiv \frac{L_{a,b}}{m} \left(\frac{1}{z} + \frac{1}{z-1} + \frac{1}{z-2} \right) \\ &\equiv \frac{L_{a,b}}{m} \frac{f_2(z)}{z(z-1)(z-2)} \equiv 0 \pmod{\ell}. \end{aligned}$$

The endpoint term $L_{a,b}/b$ is an ℓ -adic unit. In

$$X_{a,b} = \frac{L_{a,b}}{L_{a,b-1}} X_{a,b-1} + \frac{L_{a,b}}{b},$$

the first multiplier is an ℓ -adic unit, so $\ell \nmid X_{a,b-1}$. The asserted denominator decrease follows. $\square$

3 Prime products

Fix a large integer n . For every integer d with $1 \leq d \leq \sqrt{n} - 1$, let

$$S_d = S_d(n) = \left\{ p : \frac{n}{d+1} < p \leq \frac{n}{d}, f_d \text{ has a root modulo } p \right\},$$

and let $T_d = T_d(n)$ be the complementary set of primes in the same interval. Put

$$S = \bigcup_{1 \leq d \leq \sqrt{n}-1} S_d, \quad Q = \prod_{q \in S} q. \quad (7)$$

For a fixed positive integer D , put

$$T^{(D)} = \bigcup_{d=1}^D T_d, \quad P = \prod_{p \in T^{(D)}} p, \quad \tau_D = \sum_{d=1}^D \frac{1 - \delta_d}{d(d+1)}. \quad (8)$$

Lemma 8. *As $n \rightarrow \infty$,*

$$\log Q = (c + o(1))n. \quad (9)$$

For each fixed D ,

$$\log P = (\tau_D + o(1))n. \quad (10)$$

Moreover,

$$\tau_D \longrightarrow 1 - c \quad (D \rightarrow \infty). \quad (11)$$

Proof. For each fixed d , Chebotarev's theorem together with partial summation gives

$$\sum_{p \in S_d} \log p = \left(\frac{\delta_d}{d(d+1)} + o(1) \right) n.$$

Fix R . Summing this for $d \leq R$ gives the corresponding initial part of (9). All primes belonging to S_d with $d > R$ are at most $n/(R+1)$, so the prime number theorem bounds their total logarithm by

$$\sum_{p \leq n/(R+1)} \log p = O\left(\frac{n}{R}\right).$$

First let $n \rightarrow \infty$ and then $R \rightarrow \infty$. This proves (9). Equation (10) follows in the same way, now with only the fixed set $d \leq D$, and using the complementary density $1 - \delta_d$. Finally,

$$\sum_{d=1}^{\infty} \frac{1}{d(d+1)} = 1,$$

so (11) follows from (2). □

4 Auxiliary prime powers and the choice of interval lengths

Fix $\varepsilon > 0$. By Lemma 8, we may choose D so large that

$$\frac{1}{\tau_D + 2c} < \frac{1}{1+c} + \frac{\varepsilon}{2}. \quad (12)$$

Choose an even integer

$$E > D + 1. \quad (13)$$

Choose an integer $r \geq 2$ such that

$$2^r > D, \quad (14)$$

and choose distinct primes

$$\ell_1, \dots, \ell_r \equiv 1 \pmod{12}.$$

For every such prime, 3 is a quadratic residue, and

$$f_2(X) = 3X^2 - 6X + 2 = 3(X-1)^2 - 1$$

has exactly two roots modulo ℓ_i .

Lemma 9. *There are infinitely many tuples of positive integers $(e_1, \dots, e_r)$ such that, with*

$$m_i = \ell_i^{e_i},$$

we have

$$\frac{\max_i m_i}{\min_i m_i} \longrightarrow 1 \quad (15)$$

as the tuples tend to infinity.

Proof. For $2 \leq i \leq r$, put

$$\alpha_i = \frac{\log \ell_1}{\log \ell_i}.$$

By simultaneous Dirichlet approximation, for every H there is an integer $1 \leq q \leq H^{r-1}$ such that

$$\max_{2 \leq i \leq r} \|q\alpha_i\| \leq \frac{1}{H}.$$

Along a sequence with $H \rightarrow \infty$, the corresponding q are unbounded. Indeed, if they were bounded, one fixed positive integer q would occur for arbitrarily large H , forcing $q\alpha_i \in \mathbb{Z}$ for every i . This is impossible because $\log \ell_1 / \log \ell_i$ is irrational for distinct primes ℓ_1 and ℓ_i . Take $e_1 = q$ and take e_i to be the nearest integer to $q\alpha_i$. Then

$$e_i \log \ell_i - e_1 \log \ell_1 = o(1)$$

uniformly in i , which is equivalent to (15). $\square$

For one of these tuples, put

$$M = \max_i m_i, \quad n = 2M, \quad N = \prod_{i=1}^r m_i, \quad G = \prod_{i=1}^r \ell_i. \quad (16)$$

For all sufficiently large tuples,

$$2m_i \leq n < 3m_i \quad (1 \leq i \leq r), \quad (17)$$

and

$$\log N = O(\log n) = o(n). \quad (18)$$

We now form $P = P(n)$ and $Q = Q(n)$ as in (7) and (8). Since every prime occurring in PQ is larger than $\sqrt{n}$, while the primes ℓ_i are fixed, for all sufficiently large tuples the three integers N , P , and Q are pairwise coprime.

We shall eventually take

$$b = NPQY, \quad a = b - n. \quad (19)$$

For each i , the factor

$$A_i = \frac{N}{m_i} PQ$$

is invertible modulo ℓ_i . Requiring

$$A_i Y \pmod{\ell_i}$$

to be one of the two roots of f_2 gives two allowed nonzero residue classes for Y modulo ℓ_i . By the Chinese remainder theorem, these conditions define a set

$$\mathcal{A} = \mathcal{A}(n) \subseteq \mathbb{Z}/G\mathbb{Z}, \quad |\mathcal{A}| = 2^r > D. \quad (20)$$

If $Y \pmod{G}$ belongs to $\mathcal{A}$, then Lemma 7 and (17) give

$$\nu_{\ell_i}(v_{a,b}) \leq \nu_{\ell_i}(v_{a,b-1}) - 1 \quad (1 \leq i \leq r). \quad (21)$$

5 A bounded-width family of CRT solutions

The set S_E contains

$$K = |S_E| = \left(\frac{\delta_E}{E(E+1)} + o(1) \right) \frac{n}{\log n} \quad (22)$$

primes. Here $\delta_E > 0$: for example, primes splitting completely in the splitting field of f_E form a positive-density subset of the primes for which f_E has a root. Since E is even, Lemma 4 gives

$$f_E(E - X) = f_E(X).$$

For every sufficiently large $q \in S_E$, each root x is therefore accompanied by the distinct root $E - x$. Distinctness follows from (4), because equality would give $x = E/2$.

For every $q \in S_d$, we require

$$NP \frac{Q}{q} Y \equiv x_q \pmod{q}, \quad (23)$$

where x_q is a root of f_d modulo q . For $q \in S_E$, choose a paired set of roots x_q^+, x_q^- . They give two allowed residues

$$y_q^\pm \equiv x_q^\pm \left(NP \frac{Q}{q} \right)^{-1} \pmod{q}$$

for $Y \pmod{q}$. Put, in $\mathbb{F}_q$,

$$m_q = \frac{y_q^+ + y_q^-}{2}, \quad h_q = \frac{y_q^+ - y_q^-}{2}.$$

Thus $h_q \neq 0$.

Let e_q be the CRT idempotent modulo Q satisfying

$$e_q \equiv 1 \pmod{q}, \quad e_q \equiv 0 \pmod{Q/q}.$$

Choose the integer C_q with

$$C_q \equiv h_q e_q \pmod{Q}, \quad -\frac{Q}{2} < C_q < \frac{Q}{2}. \quad (24)$$

Since $Q/q \mid C_q$, there is an integer t_q such that

$$C_q = \frac{Q}{q} t_q, \quad 0 < |t_q| < \frac{q}{2}. \quad (25)$$

Choose one prime $q_0 \in S_E$. If necessary, choose one further prime of S_E and fix one of its two roots, so that an even number $2J$ of reservoir primes remains. By (22),

$$J \asymp_E \frac{n}{\log n}. \quad (26)$$

No two of these centered integers are equal: if $C_q = C_r$ for distinct primes q, r , then the common value is divisible by both Q/q and Q/r , hence by Q , contradicting (24) and $h_q \neq 0$. Order the corresponding centered integers C_q increasingly and pair adjacent terms. If the j th pair is (q_j, r_j) , define

$$D_j = C_{r_j} - C_{q_j} > 0. \quad (27)$$

Then

$$\sum_{j=1}^J D_j < Q. \quad (28)$$

Indeed, the paired intervals are disjoint and all C_q lie between $-Q/2$ and $Q/2$.

Let $W_0 \pmod{Q}$ be the CRT solution obtained by using the midpoint residue m_q at every unfixed reservoir prime, the selected fixed root at the possible extra reservoir prime, and a fixed root at every $q \in S \setminus S_E$. Choose W_0 in $[0, Q)$. For signs

$$\sigma, \eta_1, \dots, \eta_J \in \{-1, 1\},$$

put

$$W_{\sigma, \eta} = W_0 + \sigma C_{q_0} + \sum_{j=1}^J \eta_j D_j. \quad (29)$$

The opposite signs within each adjacent pair ensure that every $W_{\sigma, \eta}$ satisfies all congruences (23). By (24) and (28),

$$-\frac{3Q}{2} < W_{\sigma, \eta} < \frac{5Q}{2}. \quad (30)$$

For $\alpha \in \mathcal{A}$, let $Y_{\sigma, \eta, \alpha}$ be the unique integer satisfying

$$0 \leq Y_{\sigma, \eta, \alpha} < GQ, \quad Y_{\sigma, \eta, \alpha} \equiv W_{\sigma, \eta} \pmod{Q}, \quad Y_{\sigma, \eta, \alpha} \equiv \alpha \pmod{G}. \quad (31)$$

By (30),

$$Y_{\sigma, \eta, \alpha} = W_{\sigma, \eta} + mQ \quad (32)$$

for some

$$m \in \{-2, -1, 0, 1, \dots, G+1\}. \quad (33)$$

The important point is that the set in (33) is fixed, independently of n .

6 Anti-concentration and the controlled insoluble primes

We use the following direct corollary of the finite-field Halász inequality of Ferber, Jain, Luh and Samotij [2, Theorem 1.4].

Lemma 10 (Finite-field Halász bound). *Let p be an odd prime, let $a_1, \dots, a_s$ be nonzero elements of $\mathbb{F}_p$, and let $\epsilon_1, \dots, \epsilon_s$ be independent uniform signs. Let R be the number of ordered signed relations*

$$\pm a_i \pm a_j = 0$$

with $i \neq j$. Then

$$\sup_{x \in \mathbb{F}_p} \Pr \left(\sum_{i=1}^s \epsilon_i a_i = x \right) \ll \frac{1}{p} + \frac{1}{s^{3/2}} + \frac{R}{s^{5/2}} + e^{-s/100}. \quad (34)$$

The implied constant is absolute.

Proof. Apply [2, Theorem 1.4] with $k = 1$, vector length s , and $M = s/100$. Its hypotheses are

$$30M \leq s, \quad 80M \leq s,$$

which hold. The quantity R_1 in that theorem counts all ordered signed two-term zero sums, including repetitions. Since p is odd and the a_i are nonzero, the repeated-index relations contribute exactly $2s$, and all remaining relations are counted by R . Substitution gives (34). $\square$

Fix $p \in T_d$ with $d \leq D$. Since $p \mid P$, put

$$z_p \equiv \frac{b}{p} = N \frac{P}{p} QY \pmod{p}.$$

The multiplier $N(P/p)Q$ is nonzero modulo p . Lemma 6 shows that it is enough to avoid a set

$$\mathcal{B}_p \subseteq \mathbb{F}_p, \quad |\mathcal{B}_p| \leq d \leq D, \quad (35)$$

of bad residues for $Y \pmod{p}$.

We next estimate the concentration of the random sum in (29). For a pair (q_j, r_j) , equations (25) and (27) give

$$D_j = \frac{Q}{q_j r_j} (q_j t_{r_j} - r_j t_{q_j}). \quad (36)$$

The integer in parentheses is nonzero: reducing a hypothetical equality modulo q_j would give $q_j \mid t_{q_j}$, contrary to (25). Its absolute value is less than $q_j r_j$.

Every reservoir prime is at most n/E , whereas every $p \in T^{(D)}$ is larger than $n/(D+1)$. Since $E > D+1$, a fixed D_j can vanish modulo at most one prime in $T^{(D)}$. Indeed, two such primes would have product larger than $n^2/(D+1)^2$, while the nonzero factor in parentheses in (36) has absolute value less than n^2/E^2 .

Let

$$z(p) = |\{j : D_j \equiv 0 \pmod{p}\}|.$$

Then

$$\sum_{p \in T^{(D)}} z(p) \leq J. \quad (37)$$

Consequently, there is at most one prime p_* for which

$$z(p_*) > \frac{J}{2}. \quad (38)$$

We call it exceptional if it exists.

For a nonexceptional p , at least $J/2$ of the coefficients D_j are nonzero modulo p . Let R_p be the number of ordered signed relations

$$\pm D_j \pm D_k = 0 \pmod{p}, \quad j \neq k,$$

among these nonzero coefficients. Then

$$\sum_{\substack{p \in T^{(D)} \\ p \neq p_*}} R_p \ll_{D,E} J^2. \quad (39)$$

To prove this, fix two distinct pairs (q, r) and (u, v) , where the four letters denote distinct reservoir primes and

$$\frac{D_j}{Q} = \frac{t_r}{r} - \frac{t_q}{q}, \quad \frac{D_k}{Q} = \frac{t_v}{v} - \frac{t_u}{u}.$$

For signs $\epsilon, \epsilon' \in \{-1, 1\}$, a congruence

$$\epsilon D_j + \epsilon' D_k \equiv 0 \pmod{p}$$

implies that p divides

$$F_{\epsilon, \epsilon'} = \epsilon uv(qt_r - rt_q) + \epsilon' qr(ut_v - vt_u). \quad (40)$$

This integer is nonzero: modulo q it is congruent to

$$-\epsilon ruv t_q \not\equiv 0 \pmod{q},$$

because q, r, u, v are distinct primes and $0 < |t_q| < q/2$. Moreover, $|F_{\epsilon, \epsilon'}| = O_E(n^4)$. Since every controlled prime exceeds $n/(D+1)$, a nonzero integer of this size has only $O_D(1)$ controlled prime divisors. Summing over the $O(J^2)$ ordered signed pairs proves (39).

Choose the signs η_j independently and uniformly. For a nonexceptional p , delete the coefficients D_j that vanish modulo p and apply Lemma 10. Uniformly in p ,

$$\rho_p := \sup_{x \in \mathbb{F}_p} \Pr \left(\sum_{j=1}^J \eta_j D_j = x \right) \ll \frac{1}{p} + \frac{1}{J^{3/2}} + \frac{R_p}{J^{5/2}} + e^{-c_0 J}$$

for an absolute $c_0 > 0$. Now

$$|T^{(D)}| = O_D \left(\frac{n}{\log n} \right) = O_{D,E}(J)$$

and, since every controlled prime is larger than $n/(D+1)$,

$$\sum_{p \in T^{(D)}} \frac{1}{p} = O_D \left(\frac{1}{\log n} \right).$$

Together with (39), this gives

$$\begin{aligned} \sum_{\substack{p \in T^{(D)} \\ p \neq p_*}} \rho_p &\ll_{D,E} \frac{1}{\log n} + \frac{|T^{(D)}|}{J^{3/2}} + \frac{1}{J^{5/2}} \sum_{\substack{p \in T^{(D)} \\ p \neq p_*}} R_p + |T^{(D)}| e^{-c_0 J} \\ &\ll_{D,E} \frac{1}{\log n} + \frac{1}{\sqrt{J}} + J e^{-c_0 J} = o(1). \end{aligned}$$

Thus

$$\sum_{\substack{p \in T^{(D)} \\ p \neq p_*}} \rho_p = o(1). \quad (41)$$

Fix $p \neq p_*$, $\sigma \in \{-1, 1\}$, and $\alpha \in \mathcal{A}$. If the integer $Y_{\sigma, \eta, \alpha}$ in (31) is bad modulo p , then by (29), (32), (33), and (35), the random sum

$$\sum_{j=1}^J \eta_j D_j \pmod{p}$$

hits one of at most

$$D(G+4)$$

fixed target residues. Hence the probability of this event is at most $D(G+4)\rho_p$. There are only $2|\mathcal{A}| = 2^{r+1}$ choices of (σ, α) . By (41) and the union bound, for all sufficiently large n there is a choice of η such that

$$Y_{\sigma, \eta, \alpha} \notin \mathcal{B}_p \quad (42)$$

for every nonexceptional $p \in T^{(D)}$, every σ , and every $\alpha \in \mathcal{A}$. Fix such a choice of η .

7 The exceptional prime and the size of Y

If no exceptional prime exists, choose any $\alpha_\sigma \in \mathcal{A}$ for each $\sigma \in \{-1, 1\}$. Suppose that p_* exists. For fixed σ , the residues

$$Y_{\sigma, \eta, \alpha} \pmod{p_*}, \quad \alpha \in \mathcal{A},$$

are pairwise distinct. Indeed, two such integers have the same residue modulo Q , so their difference is kQ with $|k| < G$. If p_* divides this difference, then $p_* \nmid Q$ and $p_* > G$ for all sufficiently large n , so $k = 0$. The two integers are then equal, and their residues modulo G are equal, forcing the two values of α to be equal.

By (20) and (35),

$$|\mathcal{A}| = 2^r > D \geq |\mathcal{B}_{p_*}|.$$

Thus, for each σ , we may choose $\alpha_\sigma \in \mathcal{A}$ such that

$$Y_\sigma := Y_{\sigma, \eta, \alpha_\sigma}$$

is good modulo p_* . By (42), both Y_+ and Y_- are therefore good at every prime of $T^{(D)}$.

Modulo Q ,

$$Y_+ - Y_- \equiv 2C_{q_0}.$$

By (25), every integer in this nonzero residue class has absolute value at least Q/q_0 : indeed,

$$2C_{q_0} + kQ = \frac{Q}{q_0}(2t_{q_0} + kq_0),$$

and the integer in parentheses is nonzero. Hence

$$|Y_+ - Y_-| \geq \frac{Q}{q_0} \geq \frac{Q}{n}.$$

Both values lie in $[0, GQ)$, so at least one of them satisfies

$$\frac{Q}{n} \leq Y < GQ. \tag{43}$$

Fix such a Y and define a, b by (19).

8 Completion of the proof

The congruences imposed above imply

$$\gcd(Y, NPQ) = 1. \tag{44}$$

Indeed, the chosen residues modulo every $q \mid Q$ are nonzero by Lemma 4; the chosen residues modulo every ℓ_i are also nonzero; and for every $p \mid P$, the residue $Y = 0$ corresponds to $z_p = 0$, which belongs to the bad set and was avoided.

For every $q \in S_d$, congruence (23) says that

$$\frac{b}{q} = NP \frac{Q}{q} Y$$

is a root of f_d modulo q . Lemma 5 therefore gives

$$\nu_q(v_{a,b}) \leq \nu_q(v_{a,b-1}) - 1 \quad (q \mid Q). \tag{45}$$

For every $\ell_i \mid G$, equation (21) gives the same decrease. For every $p \mid P$, our choice of Y and Lemma 6 give

$$\nu_p(v_{a,b}) \leq \nu_p(v_{a,b-1}). \quad (46)$$

At every other prime, Lemma 3 and (44) show that a possible increase is at most the exponent of that prime in Y . Multiplying the valuation inequalities yields

$$\frac{v_{a,b}}{v_{a,b-1}} \leq \frac{Y}{GQ} < 1 \quad (47)$$

by (43). Hence

$$v_{a,b} < v_{a,b-1}.$$

It remains to estimate the gap. By (9), (10), (18), and (43),

$$\log Y = \log Q + O(\log n) = (c + o(1))n$$

and therefore

$$\begin{aligned} \log b &= \log N + \log P + \log Q + \log Y \\ &= (\tau_D + 2c + o(1))n. \end{aligned} \quad (48)$$

In particular, b is exponential in n , so

$$a = b - n = (1 - o(1))b$$

and

$$\log a = (\tau_D + 2c + o(1))n.$$

Since $b - a = n$,

$$\frac{b - a}{\log a} = \frac{1}{\tau_D + 2c} + o(1). \quad (49)$$

By (12), the right-hand side is less than

$$\frac{1}{1 + c} + \varepsilon$$

for all sufficiently large tuples in Lemma 9. There are infinitely many such tuples, and $a \rightarrow \infty$ along them. Since $b(a) \leq b$ for each constructed value of a , the asserted upper bound for the lower limit follows as well. This proves Theorem 1.

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